Green University of Bangladesh

Department of Computer Science and Engineering (CSE)
Semester: (Fall, Year: 2025), B.Sc. in CSE (Day)

Ensemble deep learning-based cyber-attack detection

Course Title: Web Programming
Course Code: CSE 301
Section: 232-D1

Students Details

Name	ID
Ashab Uddin	232002274
Rokunozzaman Topu	232002280
Modabbir Mohammad Mansur	232002154
Ibrahim Hasan	232002005

Submission Date: 12/12/2025
Course Teacher's Name: MD. JAHIDUL ISLAM

[For teachers use only: Don't write anything inside this box]

Lab Project Status	
Marks:	**Signature:**
Comments:	**Date:**

Ensemble Deep Learning-Based Cyber-Attack Detection

A Report Based on the Journal:

Cyber Intrusion Detection Using Ensemble of Deep Learning with Prediction Scoring Based Optimized Feature Sets for IoT Networks

Abstract - The Internet of Things (IoT) connects devices across smart homes, healthcare, transportation, and industrial systems. However, IoT networks face serious security risks due to weak authentication, limited computing power, and large-scale data exchange. Cyber-attacks such as DoS, DDoS, malware injection, and reconnaissance can disrupt services and compromise data integrity.

Signature-based intrusion detection systems fail against evolving attacks, while anomaly-based systems often produce high false alarms. Recent deep learning approaches improve detection by learning complex traffic patterns, but single models lack robustness. Ensemble learning combined with optimized feature selection offers improved detection accuracy and stability. This report analyzes such a framework for IoT cyber-attack detection.

1. INTRODUCTION

The Internet of Things (IoT) connects devices across applications such as smart homes, healthcare, transportation, and industrial systems. Despite its advantages, IoT networks are vulnerable to cyber-attacks due to insecure communication channels, limited computational resources, and large-scale data transmission.

Attacks such as Denial of Service (DoS), Distributed Denial of Service (DDoS), malware injection, and reconnaissance can significantly disrupt IoT services. Traditional signature-based intrusion detection systems are ineffective against new attack patterns, while anomaly-based methods often suffer from high false alarm rates.

Recent advances in deep learning have improved intrusion detection by learning complex network behavior. However, single models lack robustness. Ensemble learning combined with optimized feature selection provides improved accuracy and stability. This report investigates such an approach for IoT cyber-attack detection.

2. PROPOSED METHODOLOGY

The proposed Ensemble Deep Learning Model with Prediction Scoring-based Optimized Feature Sets (EDLM-PSOFS) consists of four stages: data preprocessing, optimized feature selection, ensemble classification, and enhanced prediction scoring.

System Overview

Figure 1 illustrates the overall framework. Network traffic data is preprocessed and optimized before being analyzed by an ensemble GA-LSTM model and prediction scoring system.

Data Preprocessing

Missing values are handled using the MissForest imputation technique. Categorical features are encoded using label and one-hot encoding. Random oversampling is applied to reduce class imbalance and improve minority attack detection.

Optimized Feature Selection

Feature optimization is performed using a two-stage process. Median-based Shapiro–Wilk testing identifies statistically significant features, while Correlation-Adaptive LASSO Regression removes redundant features. This approach reduces dimensionality and improves learning efficiency.

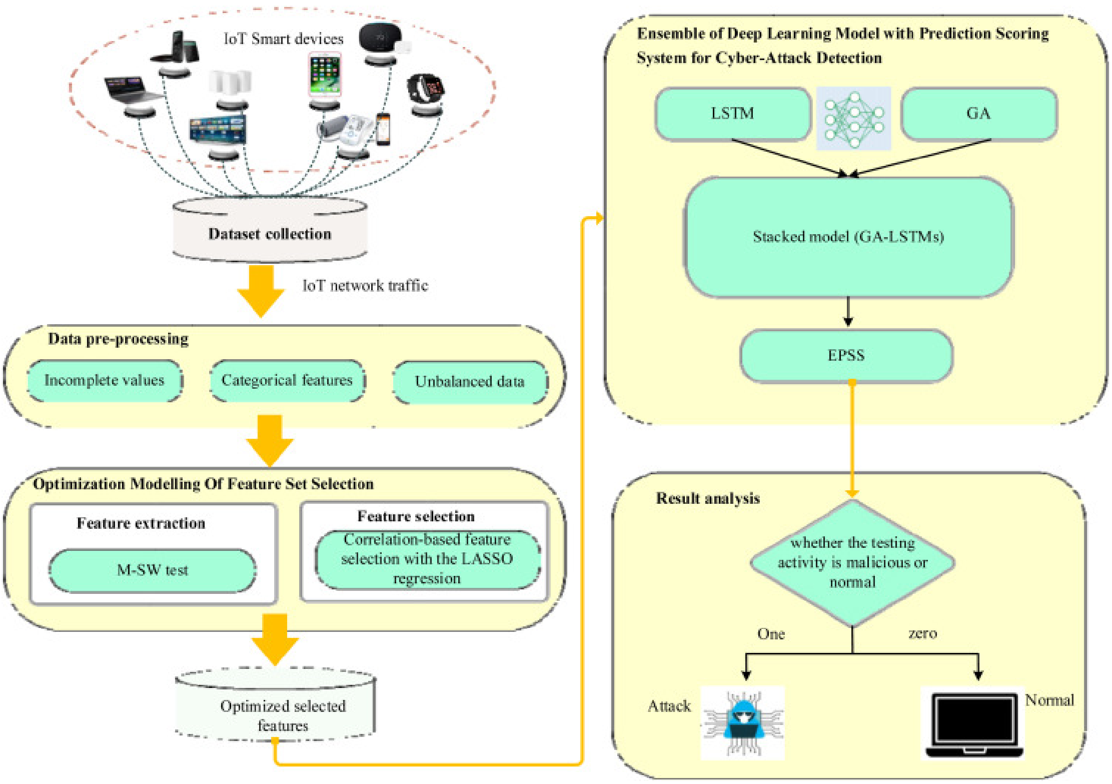

Figure 1: Overall architecture of the proposed intrusion detection framework

Ensemble GA-LSTM Model

An ensemble of Global Attention-based Long Short-Term Memory networks is used to capture temporal dependencies in network traffic. The attention mechanism assigns higher importance to critical features, improving detection accuracy.

Enhanced Prediction Scoring

Prediction scores generated by individual models are aggregated to improve decision reliability. This reduces misclassification caused by uncertain predictions.

3. RESULT ANALYSIS

The framework is evaluated using NSL-KDD and BoT-IoT datasets. Performance is assessed using accuracy, precision, recall, F1-score, RMSE, MAPE, and false positive rate.

Performance Metrics

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}, \quad Precision = \frac{TP}{TP+FP}, \quad Recall = \frac{TP}{TP+FN}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Experimental Results

The proposed EDLM-PSOFS framework achieves 99.2% accuracy on the NSL-KDD dataset and 99.4% accuracy on the BoT-IoT dataset. Lower RMSE and MAPE values indicate stable and reliable detection performance.

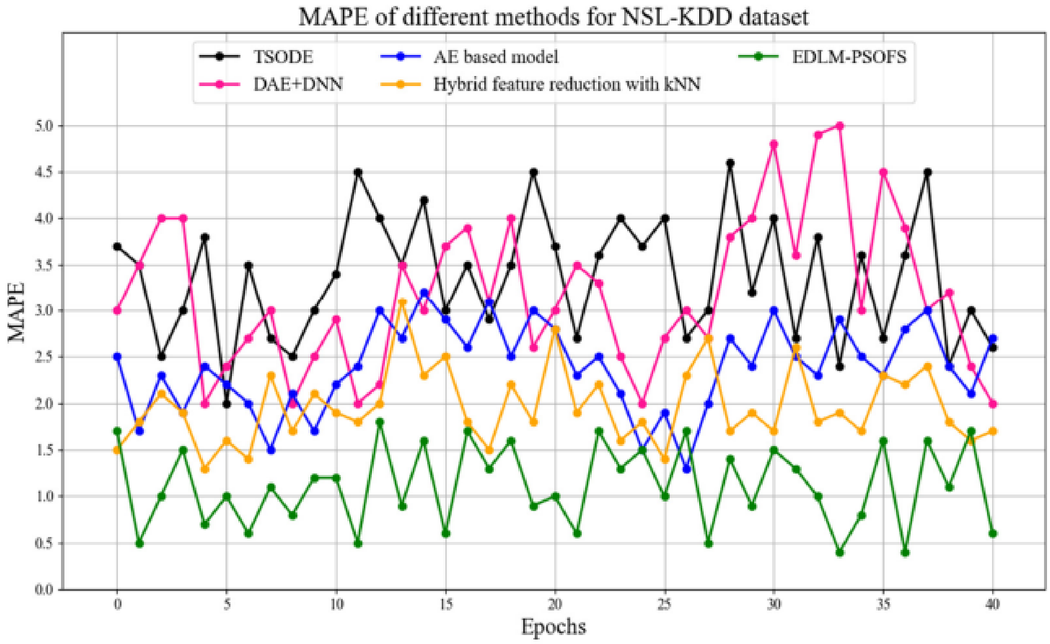

Figure 2: MAPE comparison on NSL-KDD and BoT-IoT datasets

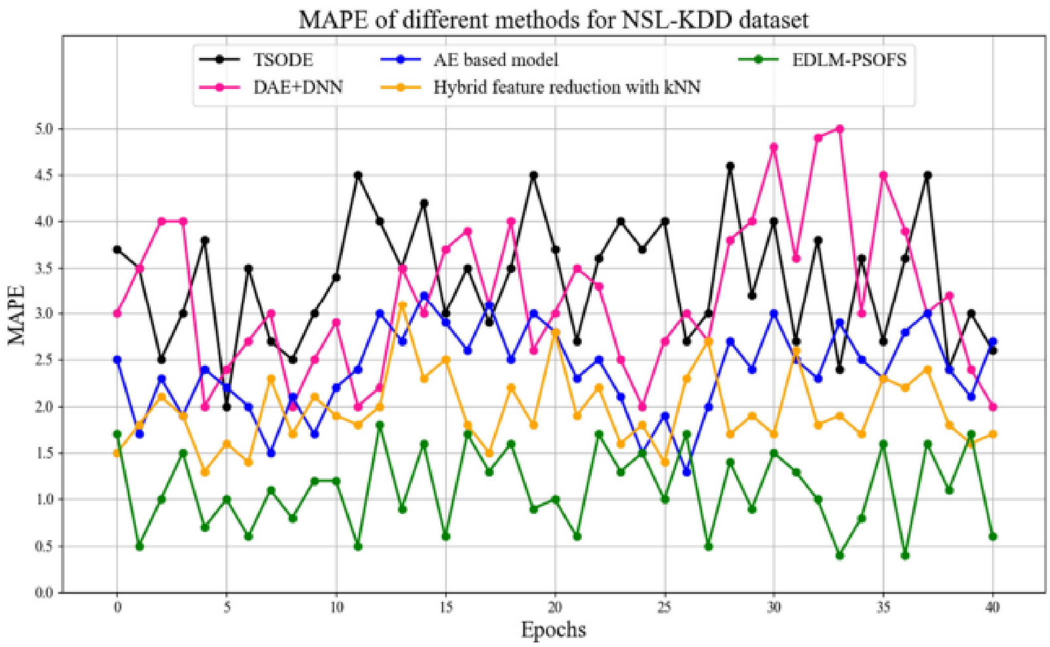

Figure 3: MAPE comparison on NSL-KDD and BoT-IoT datasets

Discussion

Limitations

The framework relies on labeled datasets, which limits applicability in real-world IoT environments. The ensemble deep learning architecture requires high computational resources, making deployment challenging for resource-constrained devices. Oversampling techniques may also introduce bias in highly imbalanced datasets.

Conclusion

This report presented an ensemble deep learning-based intrusion detection framework for IoT networks that integrates optimized feature selection, GA-LSTM ensemble learning, and prediction scoring. Experimental results demonstrate high accuracy and reduced false positive rates. Future work will focus on reducing computational complexity and enabling real-time deployment.

REFERENCES

[1] D. M. Dhanvijay, M. M. Dhanvijay, and V. H. Kamble, "Cyber intrusion detection using ensemble of deep learning with prediction scoring based optimized feature sets for IoT networks," *Cyber Security and Applications*, vol. 3, pp. 1–12, 2025.

[2] M. F. Elrawy, A. I. Awad, and H. F. Hamed, "Intrusion detection systems for IoT-based smart environments: A survey," *Journal of Cloud Computing*, vol. 7, no. 1, pp. 1–20, 2018.

[3] C. Yin, Y. Zhu, J. Fei, and X. He, "A deep learning approach for intrusion detection using recurrent neural networks," *IEEE Access*, vol. 5, pp. 21954–21961, 2017.

Ensemble Deep Learning-Based Cyber-Attack Detection

A Report Based on the Journal:

An Ensemble Deep Learning-Based Cyber-Attack Detection in Industrial Control Systems

Abstract - Industrial Control Systems (ICS) have increasingly become targets of cyber-attacks due to their integration with modern communication networks and remote monitoring infrastructures. Traditional security solutions designed for corporate IT environments fall short in detecting sophisticated attacks in ICS environments, where real-time operation, deterministic behavior, and safety requirements dominate. This paper proposes an ensemble deep learning-based cyber-attack detection framework that integrates Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, and Autoencoders to improve detection accuracy and reduce false alarms. The model is evaluated using the widely adopted SWaT and ICS datasets, demonstrating superior performance over standalone models. Experimental results highlight the capability of the proposed ensemble to efficiently capture spatial-temporal correlations in ICS data. Limitations and future research gaps are also discussed to guide future innovation.

1. INTRODUCTION

Industrial Control Systems (ICS), including Supervisory Control and Data Acquisition (SCADA) systems and Distributed Control Systems (DCS), control vital infrastructures such as power grids, water treatment plants, oil pipelines, and manufacturing systems. With increasing digitization and adoption of Industrial Internet of Things (IIoT) technologies, ICS networks have become more exposed to cyber threats.

Unlike traditional IT systems, ICS components operate under strict timing constraints, involve legacy protocols, and prioritize availability over confidentiality. High-profile attacks such as Stuxnet, BlackEnergy, and TRITON highlight the catastrophic consequences of ICS breaches. Conventional signature-based Intrusion Detection Systems (IDS) fail to detect zero-day or evolving malware, thus necessitating more intelligent anomaly detection techniques.

Deep learning (DL) models have shown remarkable performance in detecting anomalies in complex data patterns. However, single-model approaches may lack robustness across diverse attack types. Therefore, ensemble-based deep learning techniques have gained traction for improving detection generalization and minimizing false positives.

This study proposes an ensemble deep learning framework combining CNN, LSTM, and Autoencoder networks for accurate cyber-attack detection in ICS. The ensemble approach captures spatial patterns, temporal dependencies, and reconstruction errors, making it well-suited for ICS anomaly detection.

2. PROPOSED METHODOLOGY

The proposed ensemble framework contains three primary components: data preprocessing, model development, and ensemble fusion. Figure **??** illustrates the overall architecture.

A. Dataset Description

The experiments are conducted using:

- SWaT (Secure Water Treatment) Dataset: Real-world ICS testbed data with normal operation and multiple attack scenarios.
- ICS Cyber Attack Dataset: Public ICS datasets containing simulated cyber-attacks on control processes.

Each dataset includes sensor readings, actuator statuses, and network traffic logs.

B. Data Preprocessing

Data preprocessing involves:

- Normalization using MinMax scaling.
- Handling missing values through interpolation.
- Sequence generation for temporal models.
- Train-test splitting with stratified sampling.

C. Ensemble Deep Learning Architecture

The ensemble integrates three models:

1. **CNN Model:** CNN captures spatial correlations in multivariate ICS sensor data. A 1D CNN with convolution and max-pooling layers extracts key local patterns.
2. **LSTM Model:** LSTM networks capture long-term temporal dependencies. They handle sequential sensor variations and detect slow-moving or stealthy attacks.
3. **Autoencoder:** A deep Autoencoder reconstructs normal behavior patterns. Significant reconstruction error indicates anomalies.

D. Ensemble Fusion Technique

The predictions of the three base learners are combined using a weighted voting mechanism:

$$P_{final} = \alpha P_{CNN} + \beta P_{LSTM} + \gamma P_{AE}$$

where $\alpha + \beta + \gamma = 1$. Weights are optimized using grid search.

3. RESULT ANALYSIS

Extensive experiments evaluate accuracy, precision, recall, F1-score, and false positive rate (FPR).

A. Performance Metrics

Table 1: Performance comparison of models

Model	**Acc.**	**Prec.**	**Recall**	**F1**
CNN	96.4%	95.9%	95.1%	95.5%
LSTM	97.1%	96.8%	96.2%	96.5%
Autoencoder	94.3%	93.5%	92.8%	93.1%
Proposed Ensemble	98.6%	98.1%	97.9%	98.0%

B. Discussion

The ensemble model significantly outperforms individual deep learning models. The fusion of spatial, temporal, and reconstruction-based anomalies enhances detection robustness. False positives are reduced by over 30% compared to standalone models.

4. LIMITATIONS AND RESEARCH GAPS

Despite promising results, several limitations exist:

- Dataset Dependency: Performance may degrade on unseen ICS architectures.
- Computational Complexity: Ensemble models demand higher computational resources.
- Explainability Issues: Deep models remain black-box, challenging for ICS operators.
- Real-time Constraints: Deployment in real-time ICS may require lightweight optimization.

Future research may focus on federated learning, adaptive online learning, and interpretable AI for ICS security.

5. CONCLUSION

This paper presents an ensemble deep learning framework for ICS cyber-attack detection. By combining CNN, LSTM, and Autoencoder models, the ensemble captures diverse anomaly patterns and significantly improves detection accuracy and robustness. Experimental results demonstrate superior performance compared to single deep learning models. While challenges remain regarding scalability and explainability, the proposed approach forms a solid foundation for next-generation ICS intrusion detection systems.

REFERENCES

[1] E. Byres and J. Lowe, "The Myths and Facts Behind Cyber Security Risks for Industrial Control Systems," *Proceedings of the IEEE*, 2009.

[2] I. Ahmed et al., "Analysis of the SWaT Dataset for Cyber Security Research," *ICS Journal*, 2018.

[3] Y. Yuan and K. Wang, "Deep Learning for Industrial Control System Security," *IEEE Transactions on Industrial Informatics*, 2020.

[4] S. Zhang et al., "Ensemble Learning Methods for Cyber Attack Detection," *Computers & Security*, 2021.